No.	Solution	Remarks
1(a)	$a = g\sin\theta$ $= (9.81)\sin 30°$ $= 4.905$ $= 4.9 \text{ m s}^{-2}$ (shown)	[1] correct working shown
1(b)	$\sin 30° = \dfrac{h}{d}$ $d = \dfrac{0.50}{\sin 30°} = 1.0 \text{ m}$ $v^2 = u^2 + 2as$ $v^2 = 2(4.905)(1.0)$ $v = 3.13 \text{ m s}^{-1}$	[1] for correct d [1] correct equation and substitution [1] correct answer
1(c)	$s = ut + \dfrac{1}{2}at^2$ $2.0 = (3.13\sin 30)t + \dfrac{1}{2}(9.81)t^2$ $t = 0.50 \text{ s}$ horizontal distance $= 3.13\cos 30°(0.50)$ $= 1.35 \text{ m}$	[1] for correct equation and substitution [1] for correct t [1] for correct distance
1(d)(i)	In the system of object and trolley, there are <u>no external forces</u> acting on them in the horizontal direction. The only horizontal forces are contact forces (action-reaction pair) acting on each other when object hits the trolley. Hence, total momentum in the horizontal direction <u>remains constant (or is conserved)</u>.	[1] for explanation [1] for conclusion
1(d)(ii)	Applying conservation of momentum, rightward as +ve, $(0.35)(3.13\cos 30°) + (1.2)(-4.0) = (1.2 + 0.35)v_f$ $v_f = -2.48 \text{ m s}^{-1}$ final speed $= 2.48 \text{ m s}^{-1}$ Direction is to the left	[1] for correct equation and substitution [1] for correct answer [1] for correct direction
2(a)		